



TAS Faculty

Year 11 Assessment Task 2

Term 3, 2026

STUDENT NAME	CLASS
	11SOF01

TASK DETAILS	
Course Name: Software Engineering	Task Weighting: 40%
Topic: Blended Mechatronics/ OOP Practical Project with folio	Task Type: Take Home Assignment
Issue date: Week 10 – Term 2 Wednesday 24 th June 2026 – Period 1	Due date: Week 6 – Term 3 Monday 24 th August 2026 – Period 1

TASK DESCRIPTION		
Task Description: This project is to be completed in groups of 2. You must work together to define, plan, build, and test a closed-loop control system using a Microcontroller. This means the system you create must include automation and feedback using sensors. Choose 1 of the scenarios below and complete the tasks from the table below. See the marking criteria for more detail. • 2 wheeled robot that can follow a line and rescue a missing object • Servo driven grabber arm • A Simon style game • A binary clock with an alarm • Another project of your choosing which is a closed-loop control system		
Outcome	Tasks to complete	Marks
SE-11-06	a) Defines the system	/ 4
SE-11-06	b) Plan the system	/ 8
SE-11-08	c) Creates system	/ 10
SE-11-09	d) Create unit tests for system	/ 5
SE-11-08	e) Manages code and uses OOP and Git	/ 5
SE-11-09	f) Maintain a development journal	/ 8

		Total	/ 40	
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MALPRACTICE

- You must only submit work that is your own.
- Artificial Intelligence (AI) is NOT to be used for this assessment task.
- Use of AI will be considered as academic malpractice and may result in an N-Warning and/or a zero mark.

NESA TERMINOLOGY

Define: State meaning and identify essential qualities

Describe: Provide characteristics and features

Evaluate: Make judgment based on criteria; determine the value of

Identify: Recognise and name

MARKING CRITERIA

You will be assessed on how well you:

- Define the system requirements and transform those requirements into a comprehensive list of specifications and plans.
- Plan your design and provide comprehensive list of the materials requirement including the sensors, actuators, controller, frame and power sources.
- Create a fully functional mechatronics system that caters for all the requirements.
- Test and evaluate the final product and code by creating unit tests for all classes generated to cover a variety of methods and variations of parameters

OUTCOMES TO BE ASSESSED

SE-11-06	applies tools and resources to design, develop, manage and evaluate software
SE-11-08	applies language structures to refine code
SE-11-09	manages and documents the development of a software project

COMPONENT	WEIGHTING
Knowledge and understanding of course content	20%
Knowledge and skills in the practical application of the content	20%

SUBMISSION INSTRUCTIONS

When: Week 6 – Term 3,
Monday 24th August 2026.
Submitted by 10 am.

Where: All work is to be
submitted to **Microsoft
Teams**.

How: A **DIGITAL** copy of your work
MUST be submitted by 10 am on the
due date via Microsoft Teams.

All submissions must adhere to the Submission Instructions outlined here, failure to do so could mean that your task has not met the requirements.

If you have any questions or technical concerns, you must speak to your teacher prior to submission.

WHAT DO I DO IF I AM ABSENT?

If you are absent or partially absent on the day of an assessment you must:

- notify the Head Teacher of the subject by 9:00 AM with the reason for your absence.
- If a task is due that day, it can be submitted via email or delivered by someone else.

You may apply for Illness/Misadventure if:

- Illness caused a late submission (must provide a medical certificate).
- Uncontrollable events affected your performance (must provide evidence).

Complete an illness/misadventure application here:

https://online.schoolbytes.education/published_forms/7550/3f580d32-d8ce-480e-8199-2d94ab291b63

Failure to follow these procedures may result in an N-Warning and a zero mark.

Refer to the 2026 Year 12 Assessment Booklet for full details.

Marking guidelines

Task	Mark	A	B	C	D	E
Defines the system	4		4 marks Provides a detailed description of the client. Produces a detailed outline of the system being created. Produces a comprehensive list of specifications for the system.	3 marks Provides a description of the client. Produces a outline of the system being created. Produces a list of most specifications for the system.	2 marks Provides a basic description of the client. Produces a basic outline of the system being created. Produces a basic list of specifications for the system.	1 mark Produces a limited outline of the system being created.
Plan the system	8	7-8 marks Provides a comprehensive list of the materials requirement including the sensors, actuators, controller, frame and power sources. Provides accurate and detailed diagrams: • Circuit Diagram • UML Diagram • Structure chart Provides multiple algorithms which are accurate and cover complex portions of the system.	5-6 marks Provides a comprehensive list of the materials requirement. Including the sensors, actuators, controller, frame and power sources. Provides 2 complete and accurate diagrams: • Circuit Diagram • UML Diagram • Structure chart Provides 1 algorithm which is accurate and represents the system.	3-4 marks Provides a comprehensive list of the materials requirement. Including the sensors, actuators, controller, frame and power sources. Provides 1 complete and accurate diagram: • Circuit Diagram • UML Diagram • Structure chart Provides 1 algorithm which is inaccurate but attempts to represent the system.	2 marks Provides a basic list of the materials requirement which includes most of the sensors, actuators, controller, frame and power sources. Provides 1 incomplete and inaccurate diagram: • Circuit Diagram • UML Diagram • Structure chart	1 mark Provides an incomplete list of the materials requirement including the sensors, actuators, controller, frame and/or power sources.
Creates system	10	9-10 marks Creates a closed-loop control system that meets the original specifications outlined. Circuit design is safe and tidy. Portions of the system work autonomously. System is intuitive to use. System works accurately continuously. System is complete.	7-8 marks Creates a closed-loop control system that mostly meets the original specifications outlined. Circuit design is safe and mostly tidy. At least 1 aspect of the system works autonomously. System is mostly intuitive to use. System works accurately most uses. System is mostly complete.	5-6 marks Creates a closed-loop control system that somewhat meets the original specifications outlined. Circuit design is safe and somewhat tidy. System works accurately some uses.	3-4 marks Creates a system that somewhat meets the original specifications outlined. Circuit design is safe. System works sometimes.	1-2 marks Creates a system that doesn't meet the original specifications outlined.
Create unit tests for system	5	5 marks Creates unit test for all Classes generated. Unit tests cover all methods and covers variations of parameters.	4 marks Creates unit test for most Classes generated. Unit tests cover most methods and covers some variations of parameters.	3 marks Creates unit test for some Classes generated. Unit tests cover a few methods.	2 marks Creates unit test for a Classes generated. Unit tests cover at least 2 methods.	1 mark Creates a basic unit test.

Task	Mark	A	B	C	D	E
Manages code and uses OOP and Git	5	5 marks Code is clean and easy to read. Makes use of correct naming conventions, meaningful comments and spacing. Multiple commits have been made and messages are meaningful. Uses OOP design to effectively organise code into classes and objects.	4 marks Code is somewhat clean and easy to read. Mostly uses of correct naming conventions, meaningful comments and spacing. Multiple commits have been made and messages are somewhat meaningful. Uses some OOP design to attempt organising code into classes and objects.	3 marks Mostly uses of correct naming conventions, meaningful comments and/or spacing. 2 commits have been made. Attempts to use OOP design concepts.	2 marks Can include uses of correct naming conventions, meaningful comments or spacing. 1 commit has been made.	1 mark Rarely makes use of correct naming conventions, meaningful comments or spacing.
Maintain a development journal	8	7-8 marks All team members provide consistent journal entries. Maintains a complete development journal. Includes images of development which include circuits, systems and tests. Entries include insight into successes and failures.	5-6 marks All team members provide somewhat consistent journal entries. Maintains a mostly complete development journal. Includes some images of development which include circuits, systems and tests. Entries include detail of what was completed.	3-4 marks All team members provide an inconsistent journal. Maintains a somewhat complete development journal. Includes a few images of development which include circuits, systems and/or tests. Entries include some detail of what was completed.	2 marks Provides an incomplete development journal. Entries include basic detail of what was completed.	1 mark Provides an incomplete development journal with large gaps. Entries provide little detail.

FEEDBACK

Areas where you have performed well:

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Areas where you need to improve:

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Strategies to help you improve:

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